

Statistical power analysis

The code used to run the following power analyses is uploaded to <https://github.com/liobaenk/respirationAdj1/tree/main/code/analysis/poweranalysis>, and is openly accessible.

Hypothesis H1.1.1

LMM : Linearized respiratory phase alignment (PA_{LOO}) $\sim$ ITIcondition + (1 + ITIcondition | ID)

To evaluate respiratory phase adaptation, we report the Phase Locking Index (PLI) as a descriptive summary of condition-level alignment (H1.1.2). However, because metrics like the PLI aggregate trial-level variance into a single value per condition, they are reduced in statistical sensitivity required to detect subtle experimental effects. To preserve the full density of our data (320 trials per condition), our primary inferential test utilizes an LMM applied at the trial level (H1.1.1).

To do that, we quantify phase alignment by calculating the cosine of the respiratory phase at target onset for each trial relative to the circular (0 to 360°) mean phase of all other trials within that *ITIcondition* (Leave-One-Out principle; LOO), abbreviated as PA_{LOO} . This cosine-transformation linearizes the circular phase data into a continuum ranging from -1 (anti-alignment) to 1 (alignment), with an expected value of 0 under the null hypothesis of a perfectly random distribution.

Statistical power was estimated via Monte Carlo simulations using the *simr* (v1.0.9.1) and *lme4* (v2.0.1) in R (Version 4.5.0).

Anticipated effect sizes (linear units)

We evaluated four simulation scenarios (see Table 1) representing varying effect sizes ($Beta$) and levels of population heterogeneity (SD_{ID}).

- Simulation scenario 1: Beta of 0.1 for Rhythmic Fast (RhyF) and Beta of 0.06 for Rhythmic Slow (RhyS), representing moderate and subtle phase-locking effects, respectively.
- Simulation scenarios 2–3: Reduced effect sizes to identify the minimum detectable effects (MDE).
- Simulation scenario 4: Primary effect sizes (Simulation 1) tested under increased inter-individual variability.

Note that because PA_{LOO} is scaled from -1 to 1 with a null expectation of 0, increases in Beta correspond to absolute shifts in percentage toward the theoretical maximum alignment of 1.

Simulation procedure

We generated synthetic datasets with 960 trials per participant (320 per rhythmic condition: RanS, RhyS, RhyF). Statistical power was calculated through Monte Carlo simulations ($N = 500$) for the target sample size ($N = 50$). To evaluate sensitivity scaling, power curves (500 iterations per step) were generated for sample sizes ranging from $N = 50$ to $N = 80$. Residual variance was fixed at a sigma (σ) of 0.707 (corresponding to residual variance σ^2 of 0.5) to simulate a high-noise environment characteristic of trial-level respiratory dynamics.

Inference strategy

The following strategy was applied:

- Contrasts (Wald z-tests): Detection of specific PA_{LOO} differences between the random baseline (RanS) and rhythmic conditions (RhyF, RhyS).

- Power curves: Identification of the required sample size to reach a power threshold of 80% for rhythmic contrasts across the simulation scenarios.

Results

Table 1 displays parameters and results of power analysis.

Contrasts ($N = 50$)

- RanS (baseline) vs RhyF: At the target sample size of $N = 50$, power to detect the effect was high across all simulation scenarios, with the lowest power estimate for an effect size (Beta) of 0.05 in simulation scenario 3 being 87.60% (see Table 1), that means a shift towards phase alignment by approximately 5% relative to a random distribution.
- RanS (baseline) vs RhyS: At the target sample size of $N = 50$, power was above 80% only in Scenario 1, with increased population heterogeneity (Scenario 4) resulting in estimated power slightly below the standard power threshold (see Table 1). At $N = 50$, effect sizes (Beta) of 0.04 and 0.03 (Scenarios 2 and 3) are detected with a power of 68% and 44.20% respectively, signaling the need for sample size increases.

Table 1

Simulation parameters and statistical power for trial-level respiratory phase alignment analysis

Simulation scenarios	Contrast	Beta	SD _{ID}	Power [95% CI]	%Singularity
Scenario 1	RanS vs RhyF	0.1	0.1	100 [99.26, 100]	0.8
	RanS vs RhyS	0.06	0.1	96.40 [94.37, 97.85]	0.2
Scenario 2	RanS vs RhyF	0.06	0.1	96.60 [94.61, 98.01]	0.6
	RanS vs RhyS	0.04	0.1	68.00 [63.71, 72.07] [#]	0.2
Scenario 3	RanS vs RhyF	0.05	0.1	87.60 [84.39, 90.36]	0.6
	RanS vs RhyS	0.03	0.1	44.20 [39.79, 48.68] [#]	0.2
Scenario 4	RanS vs RhyF	0.1	0.15	99.60 [98.56, 99.95]	0
	RanS vs RhyS	0.06	0.15	74.60 [70.55, 78.36] [#]	0

Note. Beta = fixed effects; SD_{ID} = random-effect standard deviations (inter-individual variability), %Singularity = percentage of Monte Carlo iterations with singular fit (no other errors or warnings regarding model stability reported). In all simulations, the random-effects correlation was held at a standard 0.2 and the residual standard deviation (σ) at 0.707. Estimates are based on 500 Monte Carlo iterations per condition with $N = 50$, with a significance threshold of $\alpha = 0.05$. Confidence intervals (95%) for power estimates were calculated based on the default binomial distribution (method = Clopper-Pearson) as implemented in the used *simr* package.

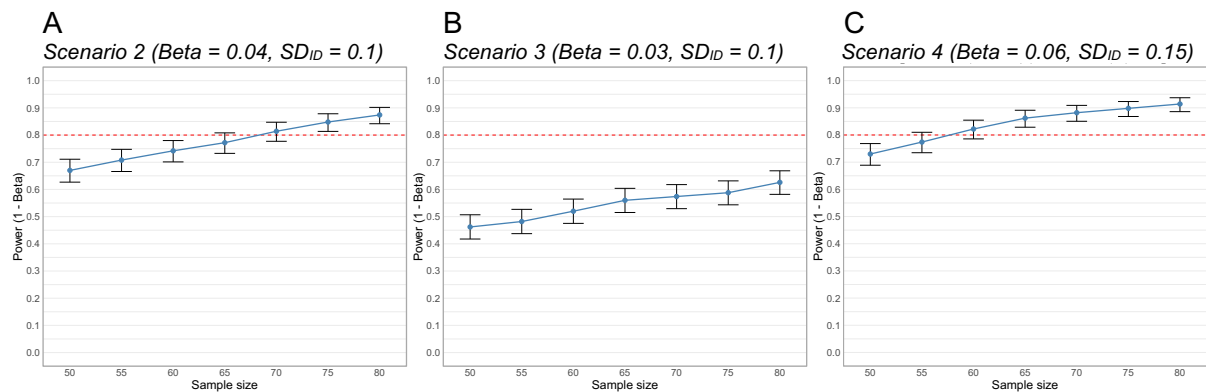
[#] Sensitivity analysis (power curves) with increasing sample size ($N > 50$) computed.

Power curves

Sensitivity analyses revealed that standard power thresholds are being reached at $N = 70$ (Power = 81.40%, 95% CI [77.71, 84.72]) to $N = 75$ (Power = 84.80%, 95% CI [81.35, 87.83]) for Scenario 2 (see Figure 1A) and $N = 60$ (Power = 82.20%, 95% CI [78.56, 85.45]) to $N = 65$ (Power = 86.20%, 95% CI [82.86, 89.10]) for Scenario 4 (see Figure 1C). Power estimates to detect an effect size (Beta) of 0.03 (Scenario 3) remain below 80% even when increasing the sample size to $N = 80$ (Power = 62.60%, 95% CI [58.19, 66.86]; see Figure 1B).

Figure 1

Sensitivity analyses (power curves) for contrast (RanS vs RhyS) on trial level



Note. Sensitivity analysis computed for contrasts with power estimates below 80% (red horizontal dotted line) at the target sample size of $N = 50$. Sample size was increased in steps of $N = 5$ up to $N = 80$. In all simulations, the random-effects correlation was held at a standard 0.2 and the residual standard deviation (σ) at 0.707. Estimates are based on 500 Monte Carlo iterations per sample size step, with a significance threshold of $\alpha = 0.05$. Confidence intervals (95%) for power estimates were calculated based on the default binomial distribution (method = Clopper-Pearson) as implemented *simr* (v1.0.9.1).

Conclusion

Based on the simulations above, we initiate data collection with a target sample size of $N = 50$, which provides power estimates above 80% for minimal detectable effects of approximately 0.05. However, should initial analyses indicate that the primary effects are between 0.04 to 0.05 or are subject to inter-individual heterogeneity above $SD_{ID} = 0.1$, we will apply a predefined stopping rule to extend data collection up to $N = 75$ (see section Stopping rule). This approach ensures the 80% power threshold is reached for these scenarios.

Hypothesis H4.1

GLMM : DetectResp ~ Signal * ITIcondition + (1 + Signal | ID)

To determine the sample size required to resolve the rhythmic modulation of behavioral detection, we conducted a simulation-based power analysis using *simr* (v1.0.9.1) and *lme4* (v2.0.1) in R (Version 4.5.0).

Model specification and parameter selection

We simulated a generalized linear mixed model (GLMM) with a probit link function to align with Signal Detection Theory (SDT). The following parameters were informed by empirical data from Forster et al. (2025):

- Baseline performance: Using sum-coding (Signal: -0.5, 0.5), the *Intercept* (representing shifts in baseline criterion, c) was set to -0.65, and the main effect of *Signal* (representing baseline sensitivity, d') was set to 1.65.
- Random effects structure: Between-subject variability was specified as 0.3 SD for the *Intercept* and 0.5 SD for the *Signal* slope.

To assess the statistical power for specific behavioral components, we utilized two model parameterizations:

1. Sensitivity (d') and criterion shifts (c) were evaluated using sum-coding (Signal: -0.5, 0.5), where a modulation of d' was tested via the interaction term.

2. Hit rate was evaluated by re-parameterizing the model with *Signal*-present as the reference level, assessing shifts in hit probability relative to the random baseline (RanS) via the main effect of *ITlcondition*.

Simulation procedure

We generated synthetic datasets with 960 trials per participant (320 per rhythmic condition, with a 3:1 *Signal* present/absent ratio). Power was calculated through Monte Carlo simulations ($N = 500$) for the sample size ($N = 50$) and via a power curve (500 iterations per step) for sample sizes ranging from $N = 50$ to $N = 80$.

Inference strategy

Statistical inference was assessed using the following metrics:

- Step A, Contrasts (Wald z-tests): Detection of specific differences in Hit rate, d' , and c (exploratory) across rhythmic conditions.
- Step B, Likelihood ratio test: First, comparing the full interaction model against a null model excluding the *ITlcondition* factor to determine the total impact of rhythmic context; second, comparing the full model against a main factor model without the interaction term (*Signal* * *ITlcondition*).

Anticipated effect sizes (probit units)

The following fixed effect sizes were informed by orientation from internal pilot data (Enk et al., 2026). As our behavioural hypotheses are undirected, predicting a general modulation of performance by rhythmic context, we employed the following effect sizes as representative theoretical benchmarks to evaluate the sensitivity of our experimental design. Note, RanS is taken as reference levels.

- Rhythmic effect 1: An interaction effect size (Beta) of 0.16, and a main effect of 0.1.
- Rhythmic effect 2: An interaction effect size (Beta) of 0.1, and a main effect of 0.06.

These values are taken to identify minimum detectable effects (MDE) and to ensure the design is scaled to resolve plausible behavioural modulations.

Results

Step A: Contrasts

- Hit rate: The design possessed 100% power (95% CI [99.26, 100.00]) to detect a modulation of 0.1.
- Sensitivity (d'): The analysis reached 82.00% power (95% CI [78.35, 85.27]) to detect an effect size (Beta) of 0.16. For a modulation of 0.1, power was 43.80% (95% CI [39.40, 48.27]); it remained below the 80% threshold even with increasing sampling (e.g., $N = 80$, power = 59.00%). The MDE resolved at $N = 50$ was identified as an effect size (Beta) of 0.15 (Power = 80.20, 95% CI [76.43, 83.61]).
- Criterion (c ; exploratory): Power to detect a shift of 0.1 was 95.00% (95% CI [92.71, 96.74]).

Step B, LRT

The LRT comparing the full interaction model to the null model (without *ITlcondition*) achieved 100% power (95% CI [99.26, 100]) at $N = 50$. Comparison of the full model against the main-factor model (without the *Signal:ITlcondition* interaction) yielded a power of 72.80% (95% CI [68.67, 76.66]) at $N = 50$.

The 80% power threshold for this interaction was reached at $N = 60$ (Power = 83.40%, 95% CI [79.84, 86.56]). Increasing the magnitude of the difference between the fixed interaction effects (i.e., *Signal:ITlconditionRhyF* vs *Signal:ITlconditionRhyS*) to 0.07 resulted in power estimates of 81.80% (95% CI [78.13, 85.09]) at $N = 50$.

Conclusion

Based on the simulation results, a sample size of $N = 50$ provides high statistical power ($\geq 95\%$) for detecting shifts in baseline hit rate and exploratory measures of response criterion shifts (c). The design is powered (82%) to detect an anticipated rhythmic modulation

of sensitivity (d') with an effect size (Beta) of 0.16. Yet, detecting a subtle modulation (Beta = 0.1) with a power of 80% would require a sample size exceeding $N = 80$. Further analysis identified the minimum detectable effect (MDE) for an interaction at $N = 50$ to be an effect size (Beta) of 0.15.

While likelihood ratio tests (LRT) confirmed that $N = 50$ is sufficient to detect the total impact of rhythmic context (full vs null model), the statistical resolution required to distinguish between specific rhythmic conditions (full vs main-factor model) is dependent on the magnitude of the interaction shift (*Signal:ITlconditionRhyS* vs *Signal:ITlconditionRhyF*). Specifically, at $N = 50$, the design is powered to resolve a difference in interaction effects of 0.07; a sensitivity to differences of 0.06 is reached at $N = 60$.

Stopping rule

To ensure robustness against smaller-than-expected effect sizes, we will implement a Bayesian sequential design (Schönbrodt et al., 2017). Data collection will proceed, at minimum, to the primary target of $N = 50$. At this checkpoint, we will evaluate the evidence for the modulation of sensitivity (d') by rhythmic context using Bayes Factors (BF_{10}). Specifically, we will test the rhythmic condition contrasts (RhyF; RhyS) against the random baseline (RanS).

Data collection will stop if the contrast for at least one rhythmic condition provides moderate evidence for either the alternative or null hypothesis ($BF_{10} > 3$ or $BF_{10} < 1/3$; van Doorn et al., 2021). If the evidence for both rhythmic contrasts remain inconclusive ($1/3 < BF_{10} < 3$), data collection will continue in steps of $N = 5$. Sampling will terminate once the evidence threshold is met, or a hard maximum of $N = 75$ is reached.

This maximum sample size ($N = 75$) represents a 50% increase over our primary target and significantly exceeds typical sample sizes of approximately $N = 40$ in recent respiratory-perceptual research (e.g., Della Penna et al., 2026; Grund et al., 2022; Johannknecht & Kayser, 2022; Kluger et al., 2021; Andrews et al., 2025, for a similar N per age group).

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